

Examiner
only

10. Urine samples from four individuals were tested in a laboratory. Some of the results are shown in **Table 10.1**. ✓ = present, ✕ = absent.

Table 10.1

Sample	Sodium chloride (salt)	Glucose	Protein
1	✓	✕	✕
2	✓	✓	✕
3	✓	✕	✓
4	✓	✓	✓

(a) (i) State the name of **one** other substance that you would expect to be present in each of the samples. [1]

(ii) Explain why you would not expect to find either protein or glucose in the urine of a healthy individual. [2]

(b) State the name of the solution you would use to test for protein in a school laboratory and state the expected colour change if sample 3 in **Table 10.1** was tested for protein. [2]



(c) Kidney failure can be treated using dialysis. **Image 10.2** shows a patient connected to a simplified drawing of a dialysis machine.

Image 10.2

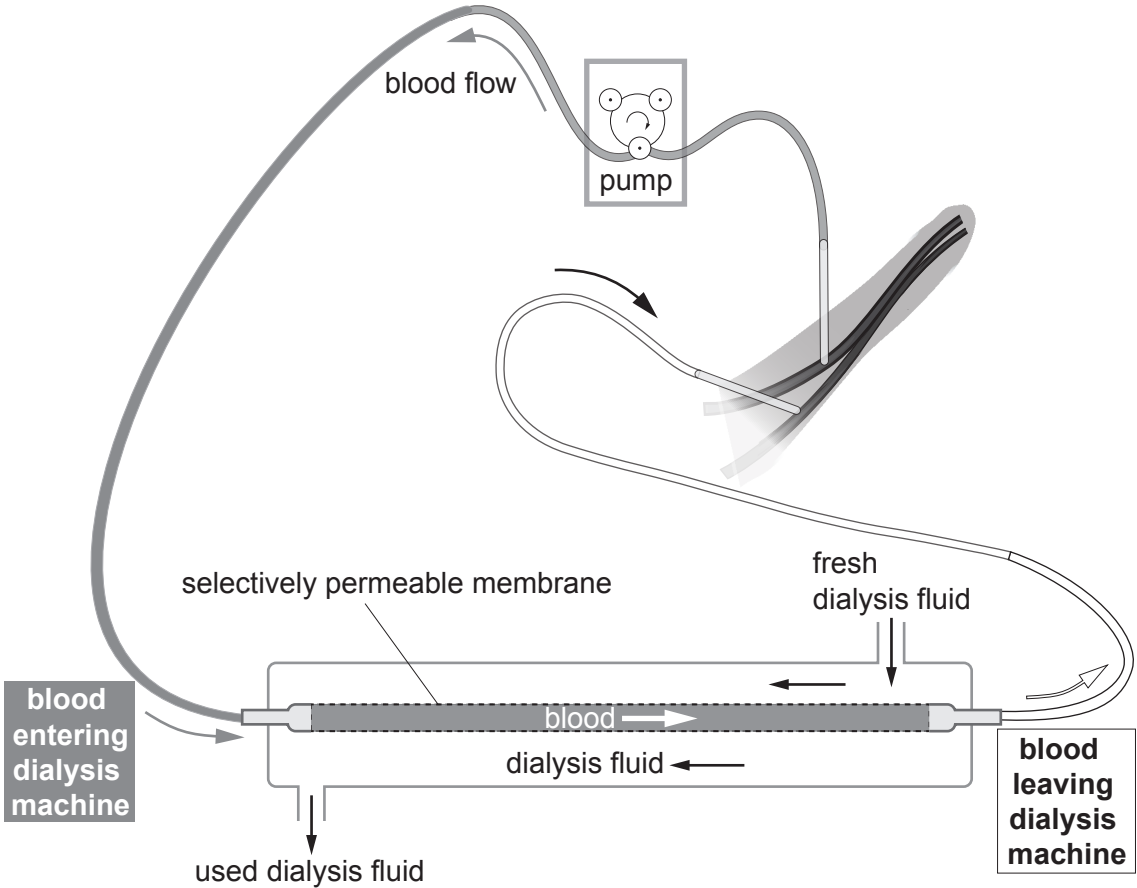


Table 10.3 compares the composition of blood and dialysis fluid.

Table 10.3

Substance	Concentration (arbitrary units)		
	Blood entering dialysis machine	Fresh dialysis fluid	Used dialysis fluid
Sodium ions	140	130	150
Chloride ions	100	90	110
Glucose	100	100	100
Urea	20	0	18
Protein	4	0	0



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- (i) The pump in the dialysis machine pumps the blood at a rate of 400 cm^3 **per minute**. Calculate the volume of blood in dm^3 that would pass through the dialysis machine if a patient was connected to the machine for **four hours**. ($1000 \text{ cm}^3 = 1 \text{ dm}^3$)

Volume of blood = dm³

- (ii) Explain why the blood and the dialysis fluid flow in opposite directions through the machine. [1]

- (iii) Using all the information given, explain how the process of dialysis takes place in the kidney machine. [6 QER]

END OF PAPER

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